

1. If $f(x) = x^2 - 2x$ and $g(x) = 8 - x^3$, evaluate the following.

a. $(f + g)(3)$ b. $(g - f)(1)$ c. $(f \cdot g)(-2)$ d. $\left(\frac{g}{f}\right)(-4)$

a. $(f + g)(3) =$ (Simplify your answer.)

b. $(g - f)(1) =$ (Simplify your answer.)

c. $(f \cdot g)(-2) =$ (Simplify your answer.)

d. $\left(\frac{g}{f}\right)(-4) =$ (Type an integer or a simplified fraction.)

2. Use $f(x) = 4x^2$ and $g(x) = \frac{x-8}{2}$ to evaluate the expression.

a. $(f \circ g)(3)$ b. $(g \circ f)(-3)$

a. $(f \circ g)(3) =$ (Simplify your answer.)

b. $(g \circ f)(-3) =$ (Simplify your answer.)

- *3. Find $f[g(x)]$ and $g[f(x)]$.

$$f(x) = \frac{x}{6} + 7, \quad g(x) = 2x - 1$$

$f[g(x)] =$

$g[f(x)] =$

- *4. Find $f[g(x)]$ and $g[f(x)]$ in the following.

$$f(x) = \sqrt{x+8}; \quad g(x) = \frac{-6}{x}$$

$f[g(x)] =$

$g[f(x)] =$

- *5. Find $f(x)$ and $g(x)$ such that $h(x) = (f \circ g)(x)$ and $g(x) = 1 - 7x$.

$$h(x) = (1 - 7x)^3 - 3(1 - 7x)^2 + 6(1 - 7x) - 1$$

$f(x) =$

6. A manufacturer of a particular item has monthly fixed costs of \$400 and variable costs of \$10 per item, and it sells the items for \$30 per item.

- a. Write a function that models the monthly profit P from the production and sale of x units of the item.
b. What is the profit if 190 items are produced and sold in 1 month?
c. At what rate does the profit grow as the number of items increases?

The function that models the monthly profit P from the production and sale of x units of the item is $P =$.
(Simplify your answer. Do not factor.)

The profit of 190 items produced and sold in 1 month is \$.
(Simplify your answer.)

The rate at which the profit grows is \$ per item.
(Simplify your answer.)

7. Suppose the weekly cost for the production and sale of bicycles is $C(x) = 20x + 3770$ dollars and that the total revenue is given by $R(x) = 85x$ dollars, where x is the number of bicycles.
- Write the equation of the function that models the weekly profit from the production and sale of x bicycles.
 - What is the profit on the production and sale of 200 bicycles?
 - Write the function that gives the average profit per bicycle.
 - What is the average profit per bicycle if 200 are produced and sold?
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a. $P(x) =$

b. The profit is \$.

c. $\bar{P}(x) =$

d. The average profit per bicycle is \$. (Round to the nearest cent as needed.)